SECURITY ASSESSMENT REPORT

Network Reconnaissance & Port Scanning Audit

Prepared for: Sandbox Security Administration | **Status:** Final Draft

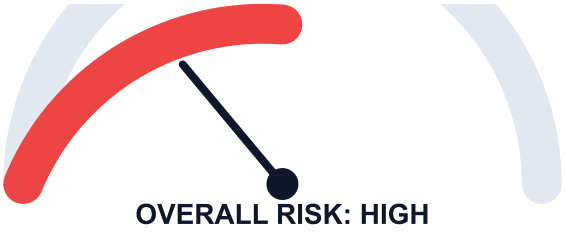
1. Executive Summary

This document outlines the findings of an authorized network vulnerability assessment and reconnaissance operation performed against the designated isolated sandbox environment. The primary objective of this engagement was to actively map the target's attack surface, enumerate running services, and analyze network traffic for insecure protocols and plaintext credential exposure.

Utilizing industry-standard methodologies, the assessment combined aggressive Nmap port scanning (TCP SYN, UDP, OS detection) with deep packet inspection via Wireshark. Critical vulnerabilities were identified, primarily stemming from the use of deprecated, unencrypted administrative protocols, which pose immediate risks to network confidentiality and integrity.

2. Target Scope & Methodology

Parameter	Detail
Target Asset	Authorized Sandbox VM (Simulated Target)
Target IP Address	192.168.56.101
Network Boundary	Internal Host-Only Virtual Network (VLAN 10)
Primary Tools	Nmap (v7.93), Wireshark (v4.0.3)
Scan Types Executed	TCP SYN (-sS), UDP (-sU), Service Versioning (-sV), OS Detect (-O)



3. Nmap Reconnaissance Results

An aggressive TCP SYN scan alongside service version detection was executed to enumerate open ports and identify the underlying daemon configurations. The output revealed a heavily exposed attack surface running legacy services.

Terminal - nmap -sS -sV -O 192.168.56.101

```
Starting Nmap 7.93 ( https://nmap.org ) Nmap scan report for 192.168.56.101 Host is up (0.00042s latency). Not shown: 995 closed tcp ports (reset) PORT STATE SERVICE VERSION 21/tcp open ftp vsftpd 2.3.4 22/tcp open ssh OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0) 23/tcp open telnet Linux telnetd 80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2) 445/tcp open netbios-s Samba smbd 3.X - 4.X (workgroup: WORKGROUP) MAC Address: 08:00:27:A3:B2:C1 (Oracle VirtualBox virtual NIC) Device type: general purpose Running: Linux 2.6.X OS CPE: cpe:/o:linux:linux_kernel:2.6 OS details: Linux 2.6.9 - 2.6.33 Nmap done: 1 IP address (1 host up) scanned in 11.34 seconds
```

Enumerated Services & Vulnerability Mapping

Port	Service	Version	Potential Attack Vector
21/TCP	FTP	vsftpd 2.3.4	Known backdoor vulnerability (CVE-2011-2523); Plaintext auth.
23/TCP	Telnet	Linux telnetd	Unencrypted management protocol susceptible to packet sniffing.
80/TCP	HTTP	Apache 2.2.8	Outdated web server version; potential CGI/directory traversal.
445/TCP	SMB	Samba smbd 3.X	Potential SMB null session enumeration or remote code execution.

4. Wireshark Traffic Analysis

Following the active network mapping, a packet capture (PCAP) session was initiated using Wireshark on the local subnet to monitor for anomalous traffic and unencrypted data transmission. Analysis immediately isolated active communication over deprecated protocols.

Packet Capture Evidence (Simulated)

No.	Time	Source	Destination	Protocol	Info
1042	14.0023	192.168.56.10	192.168.56.101	TCP	52410 → 21 [SYN] Seq=0 Win=64240 Len=0
1045	14.0531	192.168.56.101	192.168.56.10	FTP	Response: 220 (vsFTPd 2.3.4)
1068	15.2210	192.168.56.10	192.168.56.101	FTP	Request: USER admin
1072	15.8904	192.168.56.10	192.168.56.101	FTP	Request: PASS s3cr3tP@ssw0rd!
1075	16.1022	192.168.56.101	192.168.56.10	FTP	Response: 230 Login successful.
1180	18.4501	192.168.56.10	192.168.56.101	TELNET	Telnet Data ... "root"

HIGH RISK **Plaintext Credential Exposure**

As demonstrated in packets 1068 through 1072, an administrator authenticated to the vsftpd service over an unencrypted channel. The credentials (`admin` / `s3cr3tP@ssw0rd!`) were transmitted in cleartext, easily intercepted by any malicious actor performing ARP spoofing or standard passive network sniffing on the local subnet.

Identified Attack Vectors

The combination of port scanning and traffic analysis reveals two critical attack vectors: the ability to passively harvest valid network credentials via sniffing (FTP/Telnet), and the ability to actively exploit a documented backdoor (CVE-2011-2523) in the FTP daemon to gain an unauthorized root shell.

5. Remediation & Strategic Recommendations

The simulated target environment exhibits a severe lack of baseline network security controls, relying on legacy protocols that fundamentally compromise data confidentiality. Immediate action is required to harden the host operating system and modernize internal communication standards.

1. Deprecate Plaintext Protocols HIGH

Immediately disable the FTP (Port 21) and Telnet (Port 23) daemons. Any protocol that transmits authentication data in cleartext is in direct violation of modern security standards. Transition all file transfer and remote management operations to secure alternatives.

2. Implement Secure Shell (SSH) INFO

Mandate the use of SSH (Port 22) for all command-line remote administration, and SFTP (SSH File Transfer Protocol) for file transfers. Ensure SSH is configured to reject password authentication in favor of cryptographic key pairs (RSA 2048+ or Ed25519).

3. Patch Management MED

The running version of vsftpd (2.3.4) contains a well-documented malicious backdoor introduced in 2011. Additionally, Apache 2.2.8 is severely outdated. Implement a continuous vulnerability management lifecycle to ensure all public-facing daemons are running current, supported releases.

4. Host-Based Firewall Enforcement MED

Deploy a host-based firewall (e.g., `iptables` or `ufw` on Linux) configured with a strict default-deny inbound policy. Only expose ports that are absolutely necessary for business function (e.g., Ports 80 and 22), explicitly dropping all other unsolicited traffic.

Conclusion

The network reconnaissance audit successfully mapped the target's perimeter, proving that a malicious actor could easily compromise the environment using passive sniffing or automated exploit frameworks. Applying the remediation steps above will effectively neutralize these specific attack vectors and drastically improve the host's defensive posture.